

[Secure Exit](#)

CO3-AT3-Debugging and Optimization

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QUESTIONS

Q1

ATM Withdrawal – Exception
Handling Debugging

10 marks

Q2

Hospital Registration –
NullPointerException
Debugging

10 marks

Q3

E-Commerce Product Search –
Array Exception Debugging

10 marks

Q4 Bank Account – Race Condition Debugging

10 marks

A bank account contains ₹10,000. Two ATM threads simultaneously attempt to withdraw ₹7,000 and ₹6,000.

The following program contains a race condition.

Task:

1. Identify the race condition.
2. Identify the critical section.
3. Explain why both threads may pass the balance check.
4. Modify the complete program using synchronized.
5. Ensure that the balance can never become negative.
6. Submit the COMPLETE corrected Java program.



SIMATS Online Programming Lab

JAVA PROGRAMMING



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**Q5**

Railway Reservation –
Synchronization Debugging

```
int balance = 10000;
```

```
void withdraw(int amount) {
```

```
if (balance >= amount) {
```

10 marks

5/5 answered

```
System.out.println(
Thread.currentThread().getName()
+ " withdrawing " + amount);

try {
Thread.sleep(100);
}
catch (InterruptedException e) {
System.out.println("Interrupted");
}

balance = balance - amount;

System.out.println(
Thread.currentThread().getName()
+ " withdrawal successful");
}
else {
System.out.println(
Thread.currentThread().getName()
+ " insufficient balance");
}
}
}

class ATM extends Thread {

BankAccount account;
int amount;
```

```
ATM(BankAccount account, int amount) {  
    this.account = account;  
    this.amount = amount;  
}  
  
public void run() {  
    account.withdraw(amount);  
}  
}  
  
class BankApplication {  
  
    public static void main(String[] args)  
        throws InterruptedException {  
  
        BankAccount account = new BankAccount();  
  
        ATM atm1 = new ATM(account, 7000);  
        ATM atm2 = new ATM(account, 6000);  
  
        atm1.setName("ATM-1");  
        atm2.setName("ATM-2");  
  
        atm1.start();  
        atm2.start();  
  
        atm1.join();  
        atm2.join();  
  
        System.out.println(  
            "Final Balance = " + account.balance);  
        }  
    }  
}
```

Correct the program so that only one withdrawal succeeds.

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         BankAccount account = new BankAccount();
5         ATM atm1 = new ATM(account, 7000);
6         ATM atm2 = new ATM(account, 6000);
7         atm1.setName("ATM-1");
8         atm2.setName("ATM-2");
9         atm1.start();
10        atm2.start();
11        try{
12            atm1.join();
13            atm2.join();
14        }
15        catch(InterruptedException e){
16            System.out.println("Main thread Interrupted");
17        }
18        System.out.println("Final Balance = "+account.getBalance());
19    }
20 }
21 class BankAccount{
22     int balance=10000;
23     synchronized void withdraw(int amount){
24         if(balance>=amount){
```

Test Input

stdin input (optional)

Output

ATM-1 withdrawing 7000
ATM-1 withdrawal
successful
ATM-2 insufficient
balance
Final Balance = 3000

🕒 Pending manual review

```
25         System.out.println(Thread.currentThread
26
27         try{
28             Thread.sleep(100);
29         }
30         catch(InterruptedException e) {
31             System.out.println("Interrupted");
32         }
33         balance=balance-amount;
34         System.out.println(Thread.currentThread
35     }
36     else{
37         System.out.println(Thread.currentThread
38     }
39 }
40 }
41 class ATM extends Thread{
42     BankAccount account;
43     int amount;
44     ATM(BankAccount account, int amount) {
45         this.account=account;
46         this.amount=amount;
47     }
48     public void run() {
49         account.withdraw(amount);
50     }
51 }
```



✓ Submitted

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